The gene machine

***p. 63* Brains may be regarded as analogous in function to computers.**

Statements like this worry literal-minded critics. They are right, of course, that brains differ in many respects from computers. Their internal methods of working, for instance, happen to be very different from the particular kind of computers that our technology has developed. This is no way reduces the truth of my statement about their being analogous in function. Functionally, the brain plays precisely the role of on-board computer—data processing, pattern recognition, short-term and long-term data storage, operation coordination, and so on.

Whilst we are on computers, my remarks about them have become gratifyingly—or frighteningly, depending on your view—dated. I wrote (p. 62) that 'you could pack only a few hundred transistors into a skull.' Transistors today are combined in integrated circuits. The number of transistor-equivalents that you could pack into a skull today must be up in the billions. I also stated (p. 66) that computers, playing chess, had reached the standard of a good amateur. Today, chess programs that beat all but very serious players are commonplace on cheap home computers, and the best programs in the world now present a serious challenge to grand masters. Here, for instance, is the *Spectator*'s chess correspondent Raymond Keene, in the issue of 7 October 1988:

> It is still something of a sensation when a titled player is beaten by a computer, but not, perhaps, for much longer. The most dangerous metal monster so far to challenge the human brain is the quaintly named 'Deep Thought', no doubt in homage to Douglas Adams. Deep Thought's latest exploit has been to terrorise human opponents in the US Open Championship, held in August in Boston. I still do not have DT's overall rating performance to hand, which will be the acid test of its achievement in an open Swiss system competition, but I have seen a remarkably impressive win against the strong Canadian Igor Ivanov, a man who once defeated Karpov! Watch closely; this may be the future of chess.

There follows a move-by-move account of the game. This is Keene's reaction to Deep Thought's Move 22:

> A wonderful move … The idea is to centralise the queen … and this concept leads to remarkably speedy success … The startling outcome … Black's queen's wing is now utterly demolished by the queen penetration.

Ivanov's reply to this is described as:

> A desperate fling, which the computer contemptuously brushes aside … The ultimate humiliation. DT ignores the queen recapture, steering instead for a snap checkmate … Black resigns.

Not only is Deep Thought one of the world's top chess players. What I find almost more striking is the language of human consciousness that the commentator feels obliged to use. Deep Thought 'contemptuously brushes aside' Ivanov's 'desperate fling'. Deep Thought is described as 'aggressive'. Keene speaks of Ivanov as 'hoping' for some outcome, but his language shows that he would be equally happy using a word like 'hope' for Deep Thought. Personally I rather look forward to a computer program winning the world championship. Humanity needs a lesson in humility.